Project Title:

Prospectif: Data-Driven Spatial Intelligence Agent for Housing Development

Identified Issue or Challenge:

Perak is currently facing a severe property overhang, recording over 4,075 unsold housing units and ranking among the highest across Malaysian states according to Rumah.my as of June 2026. The core problem is not a lack of marketing platforms or advertising reach, but rather unstrategic location planning by developers. Residential projects are frequently built on the speculative assumption that a town's population will naturally grow, without grounding in empirical socio-economic data. Consequently, developments are placed in areas lacking economic catalysts or job creation, leaving units unsold and turning developments into stagnant overhang zones.

Proposed Solution / Innovation:

The proposed solution is Prospectif, a Predictive Site Selection AI agent engineered to guide property developers toward data-backed land selection. The platform aggregates open data from the National Property Information Centre (NAPIC) regarding overhang metrics alongside Department of Statistics Malaysia (DOSM) data on localized job growth, employment trends, and district demographics.

Powered by a spatial database (PostGIS) and a FastAPI backend, the system generates predictive heatmaps highlighting emerging boom areas driven by job creation. It provides developers with a "Feasibility Score" (1–100) for specific land plots and includes a conversational RAG AI interface, enabling stakeholders to query real demand patterns before committing to land acquisitions.

We have determined multiple variables that are important on how a city population and housing demands increasing in a city. Below are the variables:

- **Current Property Overhang (Supply Indicator):** High overhang means low viability. We need to track unsold units categorized by property type and district.
- **Employment & Labor Force (Demand Catalyst):** Job creation is the primary driver of migration. An influx of jobs in a specific district directly translates to housing demand.
- **Population Density & Migration (Demographics):** Tracking which districts in Perak are experiencing population growth.
- **Household Income & Ability to Buy a House (Supply Indicator):** Distance to existing industrial parks, highways, public transit, and commercial centers.

Target Group:

- Private property developers and Government-Linked Companies (GLCs) developer.
- Local Authorities (PBTs) and state urban planning agencies in Perak.
- Real estate investors and state housing policy formulators.

Expected Impact:

- **Mitigates Property Overhang:** Prevents future unsold inventory by ensuring housing supply directly aligns with verified employment growth and real population demand.
- **Optimizes Land & Capital Utilization:** Eliminates misallocated development capital in stagnant regions by guiding investments toward high-potential economic corridors.
- **Socio-Economic Alignment:** Ensures Perak residents gain access to housing situated near active job markets and essential urban infrastructure.

Potential for Implementation in Perak:

This innovation holds high implementation feasibility as it leverages existing open public datasets (NAPIC, OpenDOSM) and aligns directly with key focus areas outlined in the program guidelines which is "**Bandar Pintar dan Tadbir Urus**" and "**Pembangunan Ekonomi dan Daya Saing Negeri**". Perak state agencies and local councils can adopt this system as an evidence-based screening framework to evaluate and approve future housing development applications, advancing the goals of the Perak Sejahtera 2030 blueprint.

Conclusion:

Prospectif transforms Perak's real estate development strategy from speculative guesswork into data-backed spatial intelligence. By stopping speculative development trends, the system equips developers and state planners with actionable insights, fostering sustainable urban growth in line with the Perak Sejahtera 2030 vision.